

JAISP

A Joint Rubin–*Euclid* Foundation Model for Detection and Astrometry

— and a path to *Roman* —

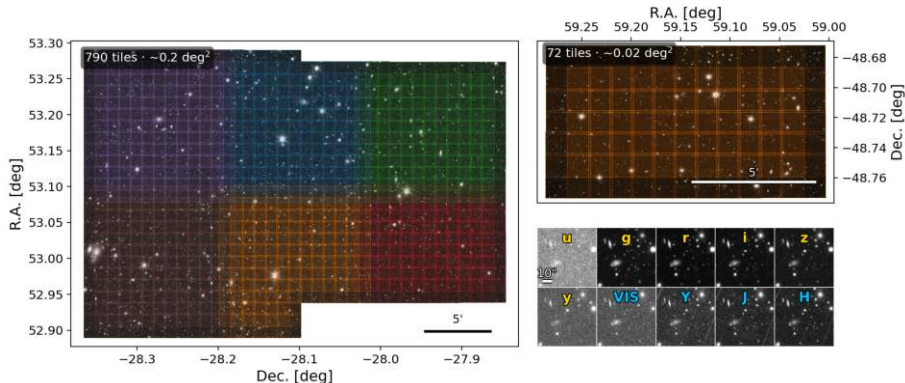
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Caltech/IPAC, NASA JPL

Shifting Landscapes in Astrophysics: New Frontiers to Explore with *Roman* · July 2026

- Cosmology is now limited by **systematics, not statistics**: whether the tensions between cosmological probes are real physics or measurement error, settling them requires control of systematics at a level no survey has yet demonstrated.
- Rubin/LSST, *Euclid*, and *Roman* will observe overlapping sky from ~ 0.3 to $2\mu\text{m}$ at sub-arcsecond resolution — combined at the pixel level, they can **cross-check one another**. But that means handling **different resolutions, PSFs, depths, and noise** at once, which no single-survey pipeline was built to do.
- AI will not replace decades of classical pipeline effort — certainly not all at once; reaching that maturity takes **comparable, problem-specific work**, and classical methods **inform the learning** along the way (here: they provide our training labels and every benchmark we test against).



- **Rubin DP1:** commissioning camera, *ugrizy*, $0.2''/\text{px}$, $\sim 1''$ seeing.
- **Euclid Q1:** VIS and near-infrared *YJH*, $0.1''/\text{px}$.
- **Training field (ECDFS):** 790 overlapping ten-band tiles over $\sim 0.2 \text{ deg}^2$.
- **Out-of-distribution field (EDF-S):** **set aside untouched** — the model never sees it — to test at the end whether everything transfers to new sky.

Self-supervised pretraining (masked-band reconstruction):

- mask one of the ten bands; reconstruct it from the other nine
- **no labels, no catalogs, no simulations**

Freeze the encoder, reuse the features:

- each task is a **lightweight head** on the same cached features
- detection & astrometry today; photometry, shapes, redshifts, ... in preparation

Cost structure

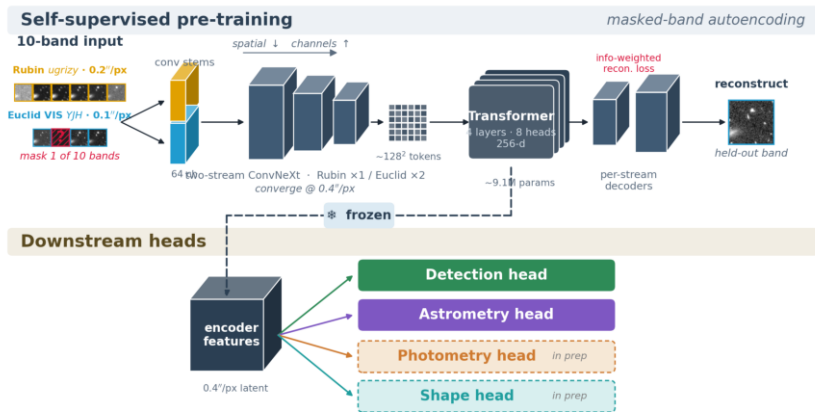
Pretraining (once): ~ 2.5
GPU-hours

Model size: ~ 9 M parameters

New sky: **one encoding pass**
(no training)

Each new task: a lightweight head

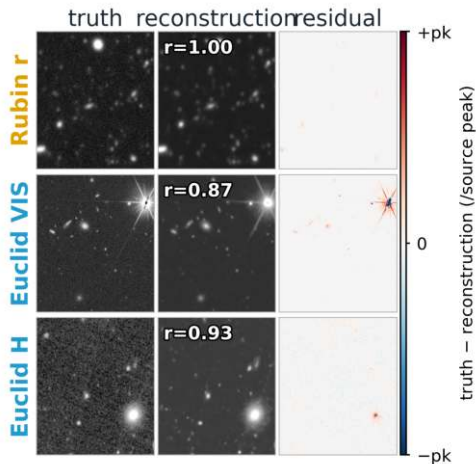
$\Rightarrow$ train once · encode each field once · measure many times



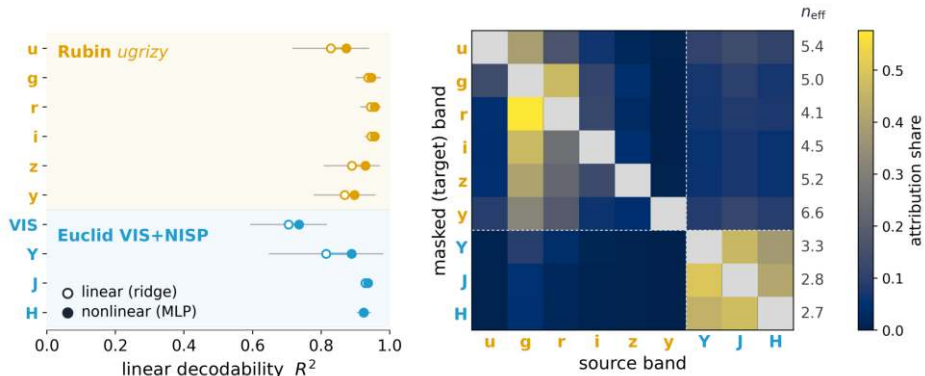
- Rubin stays at $0.2''/\text{px}$, Euclid at $0.1''/\text{px}$ — **no resampling onto a shared grid**, so neither instrument's spatial information is thrown away.
- The branches merge on a common grid, where all ten bands exchange information.

v1–v2	Contrastive patch matching — make matching Rubin/ <i>Euclid</i> cutouts look similar in feature space	failed: sky-dominated
v3	Object-centric matching — discover individual sources, pair them across instruments	abandoned: no precision
v4–v5	Feature matching on full tiles (JEPA) — make the two instruments' features agree at corresponding positions	beaten by a simple CNN
v6	The turn: predict actual pixels — hide one band, reconstruct its image; no shortcut that discards position	paradigm shift
v7–v8	Mixed resolution — one branch per instrument at its native pixel scale, merged at a common physical scale	works
v9–v10	Refinements — every band gets an equal vote in training; the loss focuses on real sources	production

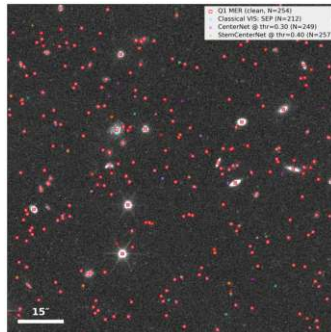
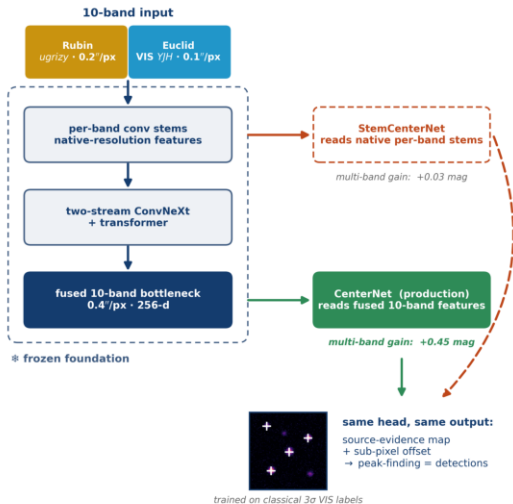
*Features that look similar aren't features that know **where** — contrastive, JEPA, masked autoencoding: we walked the road, and pixels won.*



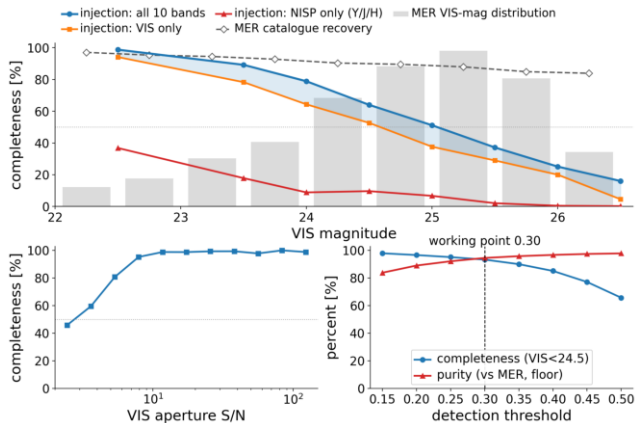
- Held-out tiles: withhold one band, predict it from the other nine.
- The loss is **noise-normalized (per-pixel variance) and weighted toward source pixels** — the model is graded on astronomical structure, not on reproducing blank sky.
- Every band is reconstructed faithfully (three examples shown) — residuals confined to bright cores and faint galaxy wings.
- A band from **one instrument predicted from bands of the other**: genuine cross-instrument correspondence, not within-instrument copying.



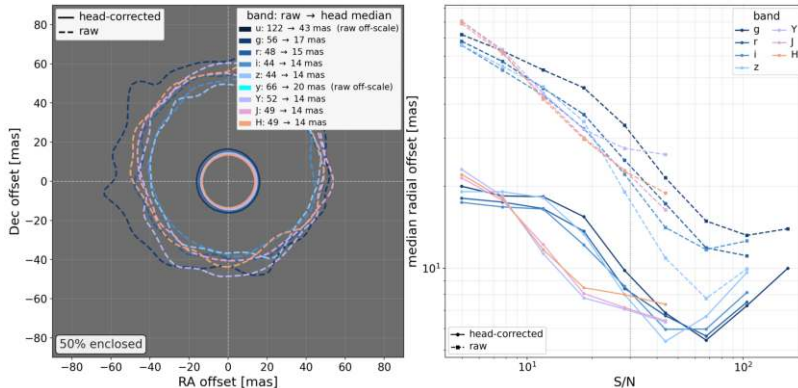
- **Left:** a simple linear fit reads each band's brightness straight back out of the features ($R^2 \approx 0.7$ – 0.95) — the information is there and easy to reach.
- **Right:** the model leans on bands the way real galaxy colors would predict: VIS is the sharp scaffold everything is built on; Rubin bands draw on their spectral neighbors; the three near-infrared bands mostly predict one another.



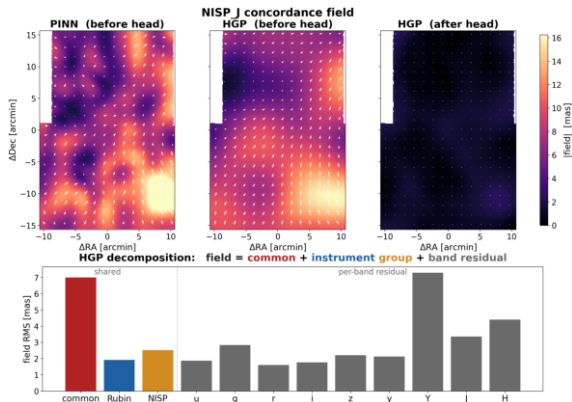
- Same head design, two read-out points: **fused ten-band features** vs. native per-band stems.
- Matches the *Euclid* Q1 MER catalog on held-out sky; **one detection per galaxy** where classical extraction fragments extended systems.



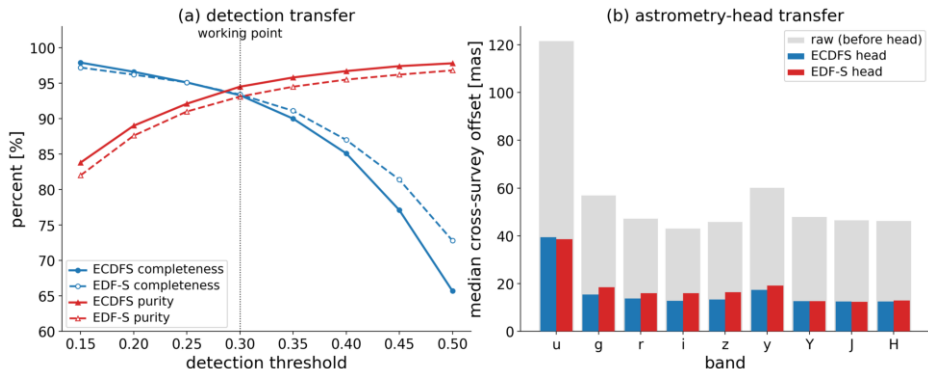
- Against the *Euclid* Q1 MER catalog on held-out sky: **93% completeness, 95% purity**.
- Injection test — real sources, dimmed to a chosen magnitude, hidden in the images: ten-band detection is **+0.45 mag deeper** than the same detector on VIS alone.
- On the deeper *Euclid* DR1 imaging, the same injection test reaches **VIS ≈ 26.0** .



- The same sources measured in both surveys disagree by ~ 50 mas — mostly photon noise in centroiding faint galaxies. A small correction network reads the model's features at each source and shrinks this to **14–17 mas in every band**.
- Ground-truth check: inject sources at **known positions** and re-measure. At $S/N = 5$ the head recovers them to **19 mas**; classical centroiding scatters by ~ 50 mas. That is within ~ 2 mas of the VIS anchor itself — **VIS-quality positions in every band**.



- Beneath the per-source scatter sits a coherent **9–10 mas pattern** between the two surveys' astrometric solutions — even though both are tied to Gaia; ~ 7 mas is shared by **all ten bands**. Two entirely different fitting methods recover the same pattern, and the per-source correction absorbs it ($\rightarrow 1.5$ mas).
- A few-mas map of **how well two independently calibrated surveys actually agree** — quality assurance no single pipeline can produce.



- The locked model and its measurement networks, applied to EDF-S: **nothing retrained**.
- Detection: 93.4%/93.1% completeness/purity vs. 93.3%/94.5% on the training field. Positions land on the same $\sim 12\text{--}18$ mas floor, within 1–3 mas.
- Scaling to more area is a matter of **applying the stack, not rebuilding it**.

Why the model is ready:

- one encoder branch per instrument *is* the design — adding *Roman* means **adding a third branch**, not redesigning the model
- training needs only **overlapping public images** — no labels, no simulations — and hours of GPU time, once WFI imaging overlaps Rubin/*Euclid* sky

What Roman adds:

- **sharp, deep near-infrared imaging at $0.11''/\text{px}$** — the infrared counterpart of the sharp VIS reference, where today we must rely on resampled NISP
- a wide survey of thousands of deg^2 **inside the LSST footprint**, plus deep time-domain fields

What Roman gains:

- detection that uses all surveys at once — for Rubin+*Euclid* that was worth $+0.45$ mag
- *Roman*-quality positions carried into the ground-based bands
- **three-way agreement maps:** Rubin–*Euclid*–*Roman* astrometry compared at the few-mas level — quality assurance no single pipeline can produce
- cross-instrument alignment solved *inside* the model instead of inherited from preprocessing (the ~ 3 mas NISP resampling error we found is exactly that)

- A small ($\sim 9\text{M}$ -parameter) foundation model, trained **without any labels** on public Rubin + *Euclid* images in ~ 2.5 GPU-hours, learns a shared ten-band representation precise enough to measure with.
- **Detection**: matches the *Euclid* catalog (93%/95%), and ten-band fusion pushes **0.45 mag deeper** than VIS alone.
- **Astrometry**: VIS-quality positions in every band — $\sim 50 \rightarrow 14\text{--}17$ mas, verified at 19 mas on hidden test sources — and a few-mas **map of the disagreement** between the two surveys' Gaia-tied solutions.
- The whole stack **transfers unchanged** to a second deep field. All of this on the *earliest* data both surveys have released — a floor, not a ceiling.
- *Roman* slots in as a **third encoder stream**: sharp NIR anchor, three-way cross-survey calibration, and one representation reused across tasks, fields — and instruments.